

Computational Wind Engineering

Finite volume method basics

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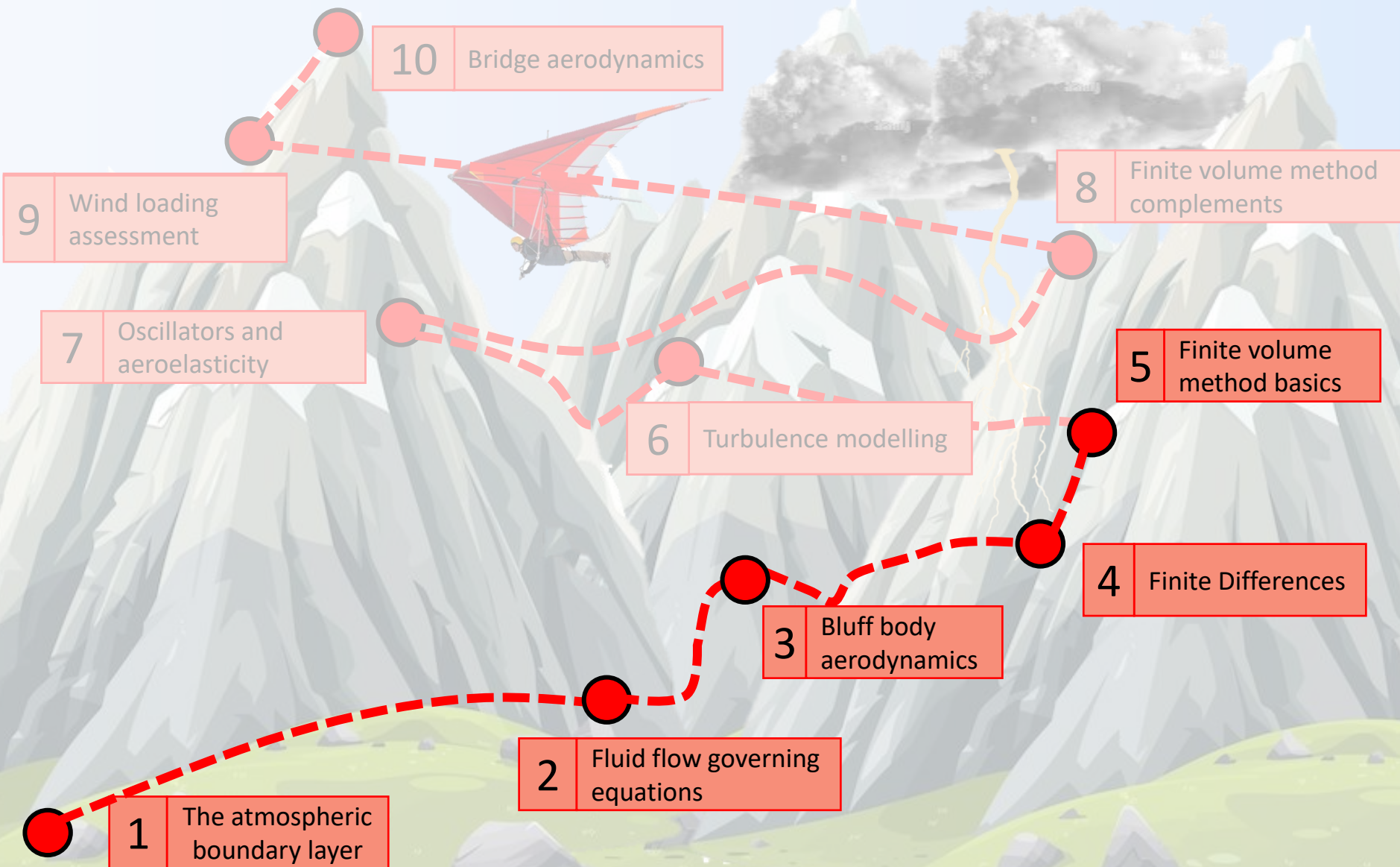
- Integrated momentum equations
- The global system
- Term by term discretization
- Grid quality
- Segregated solution

Map of the course

Wind Engineering

Fluid Dynamics

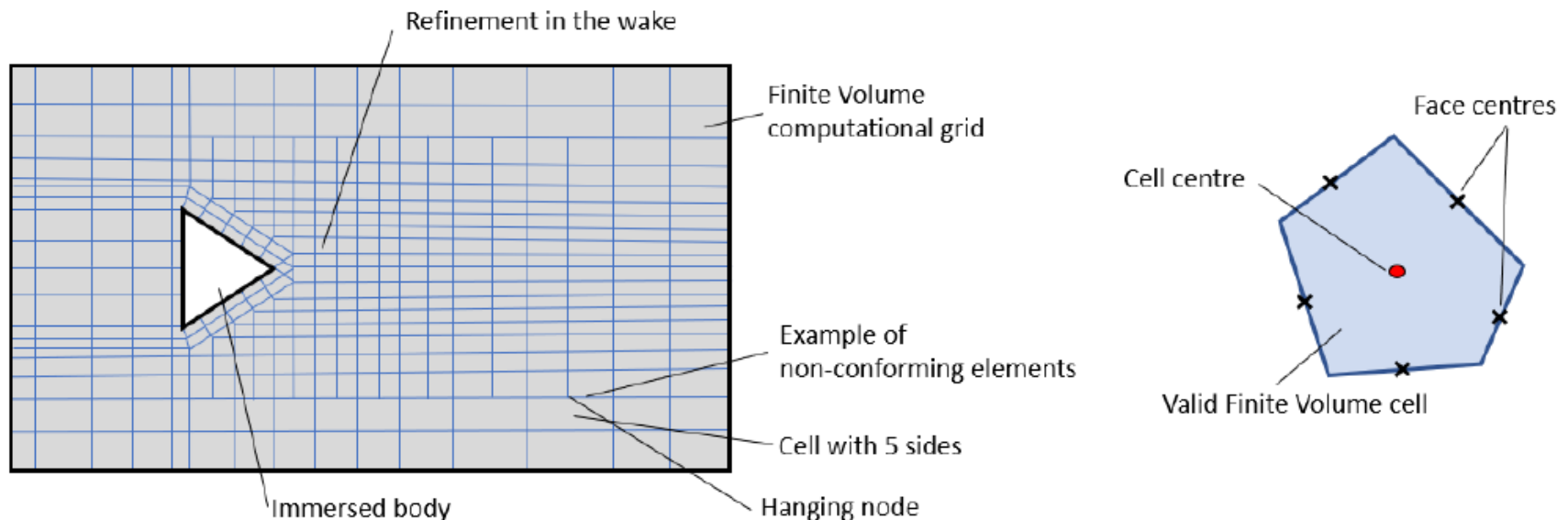
Numerical methods



The **steps needed to set a numerical analysis**, regardless of the adopted approximation strategy, are

- 1 - define the geometry of the computational domain;
- 2 - create a computational grid;
- 3 - specify the equations to be solved;
- 4 - specify the approximation schemes;
- 5 - specify boundary and initial conditions;
- 6 - solve a system of equations;

In particular, if the Finite Volume Method is chosen, we might have



To develop the method, we start from the integrated momentum balance equations. The first of these equations reads

$$\boxed{\frac{d}{dt} \int_V u dV} + \int_S u \mathbf{u}^T \mathbf{n} dS - \frac{1}{\rho} \int_S \boldsymbol{\sigma}[1, :] \mathbf{n} dS = \int_V s_{up} - s_{ud} dV,$$

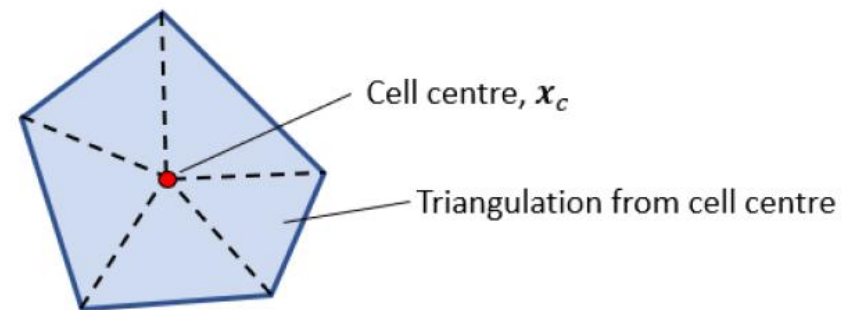
with

$$\boldsymbol{\sigma} = -p \mathbf{I} + 2\mu \nabla_s(\mathbf{u})$$

We proceed to a term by term approximation. **We need to express all terms relying only on cell centre values.** In particular, for the first term

$$\boxed{\frac{d}{dt} \int_V u dV} \approx \frac{du_c}{dt} \int_V dV = \frac{du_c}{dt} V,$$

$$\approx V_c \frac{u_c^{n+1} - u_c^n}{\Delta t}.$$



We can thus start to build a system of equation. Limiting to the unsteady terms we have

$$\begin{bmatrix} V_c/\Delta t & 0 & \dots & 0 \\ 0 & V_c/\Delta t & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & V_c/\Delta t \end{bmatrix} \begin{bmatrix} u_1^{n+1} \\ u_2^{n+1} \\ \dots \\ u_J^{n+1} \end{bmatrix} + \begin{bmatrix} -V_c/\Delta t & 0 & \dots & 0 \\ 0 & -V_c/\Delta t & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & -V_c/\Delta t \end{bmatrix} \begin{bmatrix} u_1^n \\ u_2^n \\ \dots \\ u_J^n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \dots \\ 0 \end{bmatrix},$$

which can be shortly written as

$$\underline{\mathcal{A}}_i \mathcal{U}^{n+1} + \underline{\mathcal{A}}_e \mathcal{U}^n = \mathcal{B},$$

We however take the contribution calculated from known values at the previous iteration on the right-hand side, leading to

$$\underline{\mathcal{A}}_i \mathcal{U}^{n+1} = \mathcal{B} - \underline{\mathcal{A}}_e \mathcal{U}^n,$$

All the terms detailed in the following must be assembled in such a system.

We now discretize the advective terms

$$\frac{d}{dt} \int_V u dV + \boxed{\int_S u \mathbf{u}^T \mathbf{n} dS} - \frac{1}{\rho} \int_S \boldsymbol{\varrho}[1, :] \mathbf{n} dS = \int_V s_{up} - s_{ud} dV,$$

We approximate the integral over the cell bounding surface as a **summation over the cell faces**

$$\boxed{\int_S u \mathbf{u}^T \mathbf{n} dS} = \sum_f u_f \mathbf{u}_f^T \mathbf{n}_f dS = \sum_f u_f \phi_f S_f$$

being $\phi_f = \mathbf{u}_f^T \mathbf{n}_f$ the fluxes at the cell faces. As it can be seen the problem is nonlinear, as a product between fluxes and velocity is present. **Let us now assume fluxes are known and fixed at the previous time step**

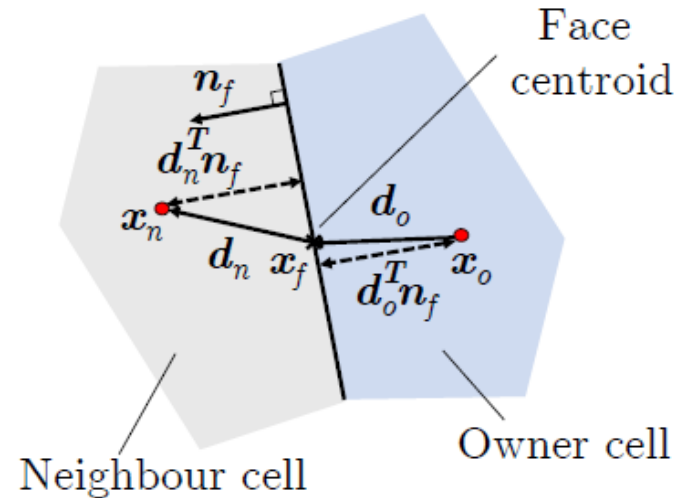
$$\int_S u \mathbf{u}^T \mathbf{n} dS \approx \sum_f u_f \phi_f^n S_f.$$

To evaluate the term we only need to evaluate u_f , the velocity at the faces, as we already did previously in the mono-dimensional case.

To evaluate the velocity at the face, starting from cell centre values

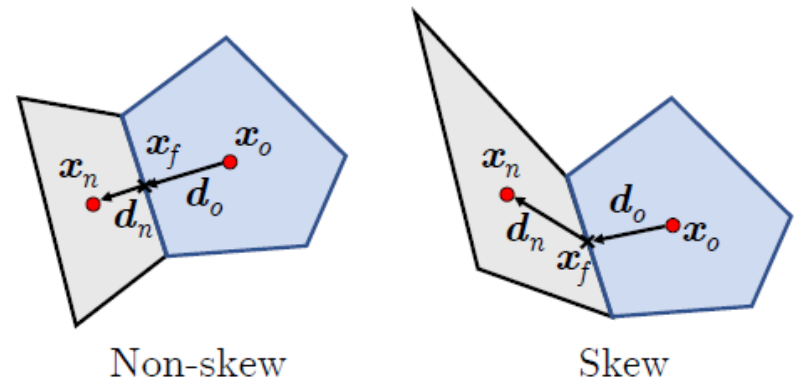
$$\begin{aligned} d_o &= x_f - x_o, & d_n &= x_n - x_f, \\ u_f &\approx \frac{n_f^T d_n}{n_f^T d_o + n_f^T d_n} u_o + \frac{n_f^T d_o}{n_f^T d_o + n_f^T d_n} u_n \\ &= \beta u_o + (1 - \beta) u_n, \end{aligned}$$

which is substantially an inverse distance weighted average.



The scheme is second order accurate and corresponds to central differentiation for regular meshes.

The approach works well for non-skew meshes, in which the line joining the cell centres contains the face centre.



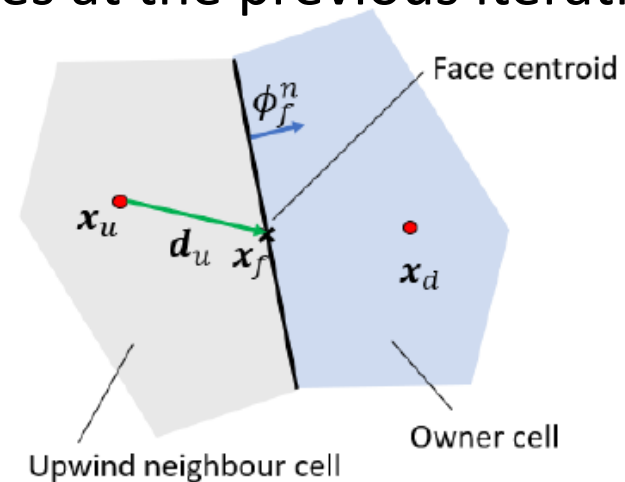
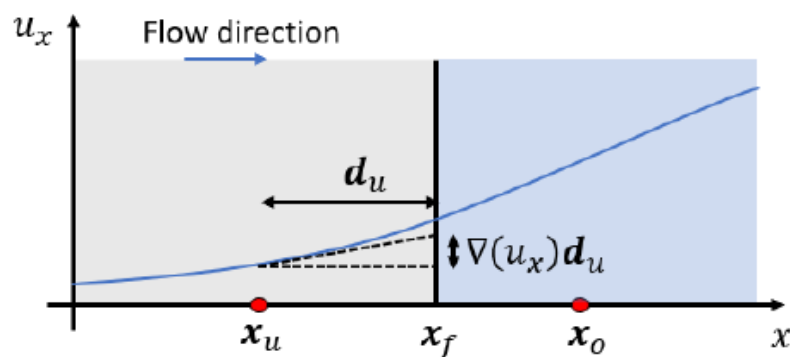
Advective terms: upwind schemes CWE

Another possibility is to consider only the upstream cell value. The resulting scheme, known as First Order Upwind, FOU, or Donor Scheme, is first order accurate. It corresponds to upwind differencing in regular meshes.

Another possibility is to use a Second Order Upwind, SOU, scheme

$$u_f = u_u + \nabla(u)|_{x_u}^n d_u$$

which, however, requires the knowledge of the velocity gradient, usually calculated explicitly, i.e. from values at the previous iteration.



The variety of schemes used to interpolate velocities from cell centres to faces is virtually unlimited, **each leading to a different compromise between accuracy, stability and computational costs.**

Another popular choice is the Linear Upwind Stabilized Transport, LUST

$$u_f = 0.75u_{f,lin} + 0.25u_{f,sou}$$

which is characterized by a good compromise between accuracy and stability.

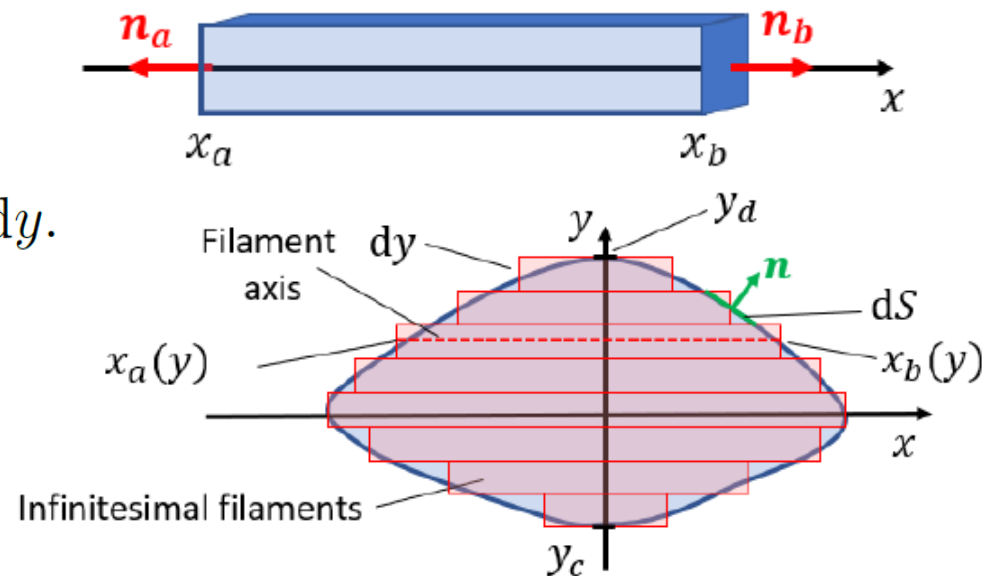
The numerical scheme used for advective terms has strong influence on the introduced numerical viscosity. In general, there is not a single optimal choice, and it is necessary to seek for a compromise between accuracy and stability.

We now need to start detailing the gradient computation, useful for instance in the SOU scheme

$$\int_{x_a}^{x_b} \frac{\partial \psi}{\partial x} dx = \psi_b - \psi_a.$$

$$\int_{y_c}^{y_d} \int_{x_a}^{x_b} \frac{\partial \psi}{\partial x} dx dy = \int_{y_c}^{y_d} (\psi_b - \psi_a) dy.$$

$$\int_V \frac{\partial \psi}{\partial x} dV = \int_{y_c}^{y_d} (\psi_b - \psi_a) |n_x| dS.$$



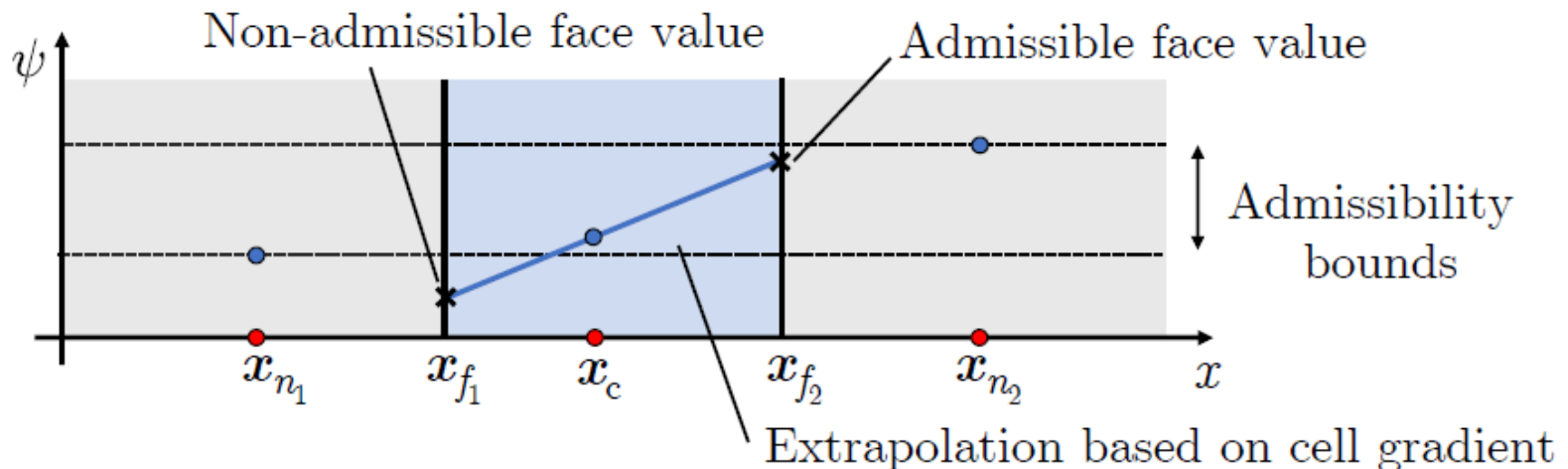
In vector form

$$\int_V \nabla(\psi) dV = \int_S \psi \mathbf{n}^T dS,$$

We can thus discretize the calculation as

$$\int_V \nabla(\psi) dV = \int_S \psi \mathbf{n}^T dS, \quad \longrightarrow \quad \nabla(\psi)_c \approx \frac{1}{V_c} \sum_f \psi_f \mathbf{n}_f^T S_f,$$

In order to preserve the stability of the simulation, gradients are often **limited** to avoid that values extrapolated at faces are outside the range defined by neighbouring cell centres.



We then consider the stress term

$$\frac{d}{dt} \int_V u dV + \int_S u \mathbf{u}^T \mathbf{n} dS - \frac{1}{\rho} \int_S \boldsymbol{\sigma}[1, :] \mathbf{n} dS = \int_V s_{up} - s_{ud} dV,$$

where

$$\int_S \boldsymbol{\sigma}[1, :] \mathbf{n} dS = - \int_S p \mathbf{I}[1, :] \mathbf{n} dS + \int_S 2\mu \nabla_s(\mathbf{u})[1, :] \mathbf{n} dS.$$

The term can be discretised as

$$\int_S \boldsymbol{\sigma}[1, :] \mathbf{n} dS \approx - \sum_f [p_f \mathbf{n}_f S_f] + \sum_f [2\mu \nabla_s(\mathbf{u})_f[1, :] \mathbf{n}_f S_f].$$

The first term requires to evaluate pressure at faces, which can be done similarly to advective terms (upwinding does not make sense in this case). The second term, after algebraic manipulation reads

$$\sum_f [2\mu \nabla_s(\mathbf{u})_f[1, :] \mathbf{n}_f S_f] = \sum_f \mu_f \nabla(\mathbf{u})_f \mathbf{n}_f S_f,$$

We thus need to evaluate the so called **normal-gradient**. To this purpose, with reference to the figure, we write

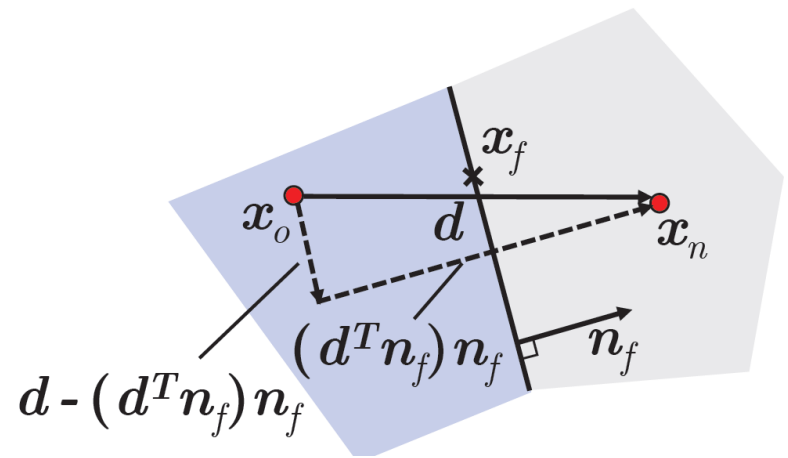
$$u_n - u_o \approx \nabla(u)_f d = \nabla(u)_f [(d^T n_f) n_f + (d - (d^T n_f) n_f)]$$

which can be rewritten as

$$\frac{u_n - u_o}{d^T n_f} \approx \nabla(u)_f \left[n_f + \left(\frac{d}{d^T n_f} - n_f \right) \right]$$

Allowing to identify

$$\nabla(u)_f n_f \approx \frac{u_n - u_o}{d^T n_f} - \nabla(u)_f \left(\frac{d}{d^T n_f} - n_f \right)$$



where the last term is calculated explicitly basing on the gradient values known at the previous iteration. **This term vanishes when d and n_f are parallel.**

The source term is treated simply as

$$\frac{d}{dt} \int_V u dV + \int_S u \mathbf{u}^T \mathbf{n} dS - \frac{1}{\rho} \int_S \boldsymbol{\varrho}[1, :] \mathbf{n} dS = \int_V s_{up} - s_{ud} dV,$$

$$\int_V s_{up} - s_{ud} dV \approx V_c (s_{up}^n - s_{ud}^{n+1}),$$

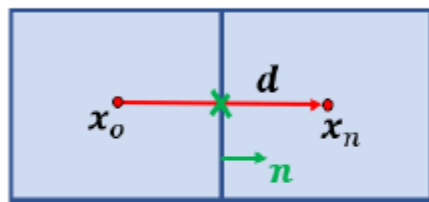
where it can be seen that sources can be evaluated explicitly or implicitly, but we do not further deepen this aspect here .

With all the aforementioned approximations, it is possible to populate the system

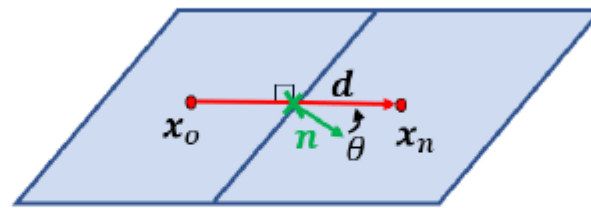
$$\underline{\mathcal{A}}_i \mathcal{U}^{n+1} + \underline{\mathcal{A}}_e \mathcal{U}^n = \mathcal{B},$$

The approximations above heavily rely on:

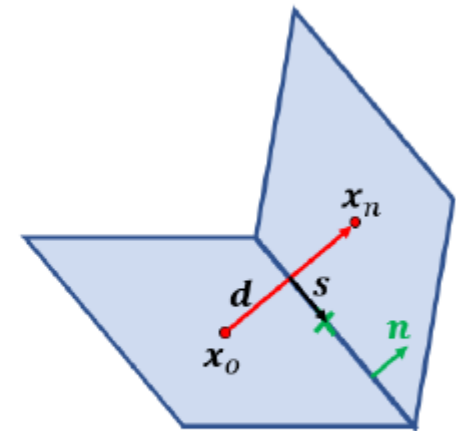
1. the evaluation of face centroid values basing on cell centre values for the advection term and pressure terms,
2. the evaluation of the face normal-gradient at the face centroid for the Laplacian term.



Orthogonal grid



Non-orthogonal grid



Skew grid

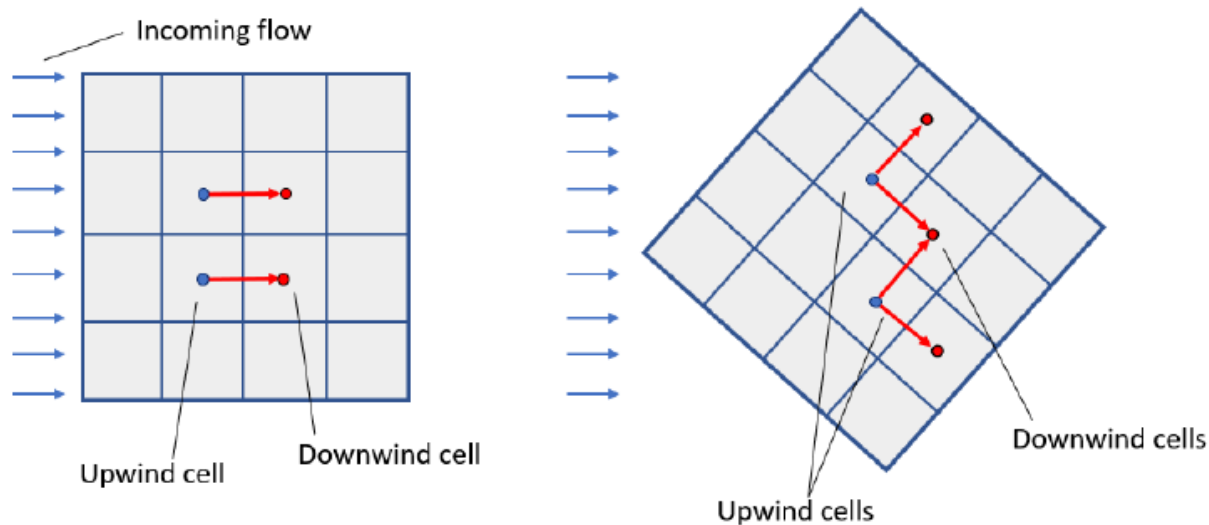
The **grid quality** is usually assessed by two indices

non-orthogonality, θ

skewness as $|s|/|d|$

which shall be kept as small as possible. They are both null for orthogonal grids.

Another factor to be considered is that numerical diffusivity increases if the flow is skew-aligned with respect to the grid.



These effects introduce additional diffusion and thus, viscosity, so that

$$\mu_e = \mu + \mu_t + \mu_{num} + \mu_{grid},$$

It can be easily understood that the effective viscosity μ_e is indeed the composition of numerous effects.

We now have to solve the resulting system. We notice that

- 1 - the evaluation of the transient term involves only degrees of freedom related to u , i.e. it is independent of v and w ;
- 2 - the advective term is nonlinear and requires all velocity components for its evaluation. However, in its discretization we used fluxes at faces calculated at the previous time step, so that the implicit evaluation of the term only requires degrees of freedom related to u ;
- 3 - the stress term has been split into pressure and viscous terms. The first one will require further discussion but we already know that the second one lead to the velocity laplacian, which requires only degrees of freedom related to u for its evaluation;
- 4 - the source term might generally depend from all degrees of freedom but we might treat explicitly only the contributions which require only degrees of freedom related to u for their evaluation.

The resulting system has the form

$$\begin{array}{l} \text{Momentum balance} \rightarrow \\ \text{Mass balance} \rightarrow \end{array} \left\{ \begin{array}{c} \left[\begin{array}{ccc|c} \times & & & \times \\ & \times & & \times \\ & & \times & \times \\ \hline \times & \times & \times & \end{array} \right] \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \\ \mathbf{w} \\ \mathbf{p} \end{bmatrix} = \begin{bmatrix} \mathbf{B}_u \\ \mathbf{B}_v \\ \mathbf{B}_w \\ \mathbf{B}_p \end{bmatrix} \end{array} \right.$$

The momentum equations are decoupled for all velocity components, being the pressure term the one coupling all the equations.

As it is much more convenient to separate the solution of each velocity component, segregated solution schemes are usually adopted.

Essentially, the solution proceeds as

1. solve all momentum equations separately, obtaining **intermediate velocities**;
2. solve for a pressure equation, which accounts for mass conservation;
3. correct **intermediate velocities** to account for mass conservation.

These operations are performed within specialised algorithms, such as **SIMPLE** (Semi-Implicit Method for Pressure-Linked Equations), **PISO** (Pressure Implicit by Splitting of Operators), etc.